

AI STUDYBUDDY

SHORT PROJECT DOCUMENTATION

1. Abstract

AI StudyBuddy is an AI-powered learning assistance platform developed to simplify and improve the study process for students. The application uses modern web technologies and Generative Artificial Intelligence to provide intelligent learning support. Students can upload study materials and generate concise summaries, flashcards, quizzes, and personalized study plans. The backend is developed using Node.js and Express.js following a RESTful API architecture. MongoDB is used to store application data, while Generative AI analyzes study materials and generates useful learning content.

2. Introduction

Students often spend considerable time reading lengthy materials, preparing notes, creating questions, and planning study schedules. AI StudyBuddy addresses these challenges by providing multiple AI-based learning features in a single platform. The main objective is to help students learn efficiently by automatically converting study materials into easy-to-understand summaries, revision flashcards, quizzes, and structured study plans.

3. Objectives

- Provide an AI-powered learning assistant.
- Generate summaries from study materials.
- Create flashcards for quick revision.
- Generate quizzes for self-assessment.
- Provide personalized study plans.
- Reduce manual preparation time.
- Provide an organized and interactive learning platform.

4. System Specifications

Hardware: Intel Core i3 or above, minimum 4 GB RAM, minimum 10 GB storage, and an internet connection.

Software: Windows/Linux/macOS, HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Generative AI API, Visual Studio Code, and Google Chrome/Microsoft Edge.

5. Proposed System

AI StudyBuddy provides an integrated platform for AI-assisted learning. Students upload study material through the application, and the backend processes the request and communicates with the Generative AI service.

AI Summarization converts lengthy study material into concise points. Flashcard Generation creates question-and-answer cards for revision. Quiz Generation creates questions based on the study material for self-assessment. Personalized Study Plan creates an organized schedule according to the student's learning requirements.

6. System Workflow

Student → Upload Material → Backend → AI Processing → Generated Content → Student

The student uploads or enters study material. The frontend sends the request to the backend, where Node.js and Express.js process it. The content is sent to the Generative AI service for analysis. The generated summary, flashcards, quizzes, or study plan is then returned to the student. Required data can be stored in MongoDB.

7. Technologies Used

Node.js – Server-side JavaScript runtime.

Express.js – Used for REST APIs and backend services.

MongoDB – NoSQL database for storing application data.

Generative AI – Used for intelligent content generation.

HTML, CSS and JavaScript – Used to develop the user interface.

8. Advantages

- Saves students' time.
- Provides quick and concise study materials.
- Supports effective revision.
- Automatically generates quizzes and flashcards.
- Helps organize study schedules.
- Provides multiple learning features in one platform.
- Offers an interactive and personalized learning experience.

9. Conclusion

AI StudyBuddy demonstrates the practical use of Generative Artificial Intelligence in education. By combining AI with modern web technologies such as Node.js, Express.js, and MongoDB, the system provides useful tools for summarization, revision, self-assessment, and study planning. Future enhancements can include AI chatbot support, voice-based learning, multilingual support, progress tracking, performance analytics, and mobile application support.